

Surface-Prior Selection for Training-Free GUI Reliability at Matched Coverage

Anonymous ACL submission

Abstract

GUI agents are usually evaluated as predictors, but deployment risk depends on which wrong actions a wrapper lets through. We study a training-free reliability wrapper around a frozen Qwen2.5-VL candidate pool and ask whether it can delay high-cost wrong executions without using calibration data, fitted thresholds, or parameter updates. The central device is matched coverage: for every benchmark family and coverage level, all selectors admit the same number of rows before official or reference scorers measure action correctness. We propose a surface-prior selector that ranks by fixed action-cost labels and breaks tied rows with public task-surface cues such as absent web candidate-set size, no previous actions, short instructions, and text/icon grounding actions. On 300-row active slices from ScreenSpot, GUIOdyssey, MMBench-GUI, and Mind2Web, this selector reduces aggregate high-cost wrong-execution area from 0.0125 for Ask-before-act to 0.0115, with paired baseline-minus-ours delta 0.0010 and 95% CI [0.0003, 0.0017]. The result is useful but narrow: family-wise improvement appears on one of four active families, task recovery is not established, and Mind2Web action F1 must remain separate from exact step success.

1 Introduction

GUI agents increasingly operate directly on screenshots, mobile screens, web pages, and desktop interfaces. This shift is visible in screenshot-based grounding models such as SeeClick and CogAgent (Cheng et al., 2024; Hong et al., 2023), high-resolution professional GUI benchmarks such as ScreenSpot-Pro (Li et al., 2025), coordinate-free and parser-based grounding systems such as GUI-Actor and OmniParser (Wu et al., 2025; Lu et al., 2024b), and broader UI understanding models such as ScreenAI and Ferret-UI (Baechler et al., 2024; You et al., 2024). These systems make GUI control

look like an ordinary prediction problem, but the evaluation problem changes once predictions become actions. A wrong click on a harmless menu, a wrong click on a destructive confirmation button, and a partially correct web action are different reliability events even when they are represented as one row in a benchmark table.

The natural response is to wrap a frozen agent with a selective execution rule: execute when the action looks safe and abstain or ask when it looks risky. This strategy connects GUI automation to older reject-option classification (Chow, 1970; Geifman and El-Yaniv, 2017), calibration and confidence estimation (Guo et al., 2017; Minderer et al., 2021), and more recent conformal or risk-controlling prediction sets (Angelopoulos and Bates, 2021; Bates et al., 2024; Angelopoulos et al., 2024). In GUI agents, however, a wrapper can look safer for reasons that do not establish safer execution. It may simply execute fewer examples, rank easy rows first, report action overlap instead of exact step success, or compare against a published baseline outside that baseline’s compatible task family. A reviewer then cannot tell whether the safety claim comes from better ordering of risky actions, lower coverage, a different scorer, or an inapplicable comparison.

This paper asks a narrower question: under identical coverage, can a training-free wrapper around a frozen GUI candidate pool reduce high-cost wrong executions relative to a strong non-learned baseline? The evaluated system sits outside Qwen2.5-VL (Qwen Team, 2025); it trains no model, uses no calibration set, fits no threshold, and only reorders fixed candidate records. Matched coverage is the central methodological device. Within each benchmark family and coverage target, every method executes the same number of rows, so abstention itself cannot be the hidden intervention. The method uses action-cost labels as the main ordering signal and breaks otherwise tied rows with a tiny surface

prior derived from task metadata. The current evidence supports an aggregate matched-coverage gain, while rejecting the stronger claim that the gain generalizes across most benchmark families.

The main intuition is deliberately modest. In a frozen candidate pool, many rows receive the same coarse risk ordering, so a selector needs a deterministic way to decide which tied actions enter the admitted set first. Public task-surface cues are not a learned safety model, but they can encode useful priors about when a wrong action is easier to inspect or less likely to be an irreversible state change. This makes the method different from both calibration-heavy guardrails and prompt-only wrappers: it is a fixed ordering rule whose value can be tested only if the evaluation holds coverage constant and separates scorer semantics. The scientific question is therefore not whether a new GUI model is stronger, but whether a tiny, auditable prior changes the order in which costly mistakes become executable.

This framing also explains why the paper keeps rejected claims in the analysis. A single aggregate win would be misleading if it were reported as desktop task recovery, robust family-wise safety, or compatibility with every published visual baseline. The audit is designed to make those boundaries visible. It reports the positive aggregate comparison, then records where the same evidence does not travel: OSWorld-style task execution is absent, Mind2Web action F1 is not exact completion, and the family-wise pattern is mostly ties. The resulting contribution is a scoped selector plus an evaluation standard for future wrapper papers.

We make four contributions. First, we define a matched-coverage protocol for high-cost wrong executions so that abstaining more cannot masquerade as safer execution. Second, we evaluate the protocol on ScreenSpot, GUIOdyssey, MMBench-GUI, and Mind2Web using existing public benchmarks and official or reference scorers. Third, we show that the surface-prior selector improves the aggregate matched-coverage area: 0.0115 versus 0.0125 for Ask-before-act, with a paired bootstrap interval entirely above zero for baseline-minus-ours. Fourth, we audit the limits of that result: task-recovery evidence is absent, family-wise improvement appears on only one of four active families, Mind2Web action F1 is not exact step success, and published GUI-Actor/OmniParser comparisons are valid only on compatible visual-family subsets.

2 Related Work

GUI grounding and visual GUI agents. Screenshot-only GUI control was accelerated by SeeClick, which tied GUI grounding pre-training to downstream GUI agent behavior and introduced ScreenSpot as a realistic grounding benchmark (Cheng et al., 2024). CogAgent, Qwen-VL variants, and ScreenAI illustrate the broader movement from OCR-like UI parsing to vision-language models that can reason over text-rich screen layouts (Hong et al., 2023; Qwen Team, 2025; Baechler et al., 2024). Later work broadened the stress tests: ScreenSpot-Pro targets high-resolution professional interfaces (Li et al., 2025); GUI-Actor replaces coordinate generation with region grounding and a verifier (Wu et al., 2025); OmniParser provides a pure-vision parsing route for GUI agents (Lu et al., 2024b); and Ferret-UI studies grounded mobile UI understanding across platforms (You et al., 2024). These papers justify evaluating frozen visual candidates, but they do not by themselves establish that a wrapper avoids high-cost wrong executions at matched coverage.

Mobile, web, and desktop agent benchmarks. Mind2Web moved web-agent evaluation toward realistic websites and open-ended instructions (Deng et al., 2023), while WebArena, VisualWebArena, and WebVoyager emphasize interactive web execution and multimodal web tasks (Zhou et al., 2024; Koh et al., 2024; He et al., 2024). Earlier and adjacent web-control benchmarks such as MiniWoB, DOM-Q-NET, and WebShop helped define the lineage from controlled web environments to open web shopping and browsing tasks (Jia et al., 2019; Yao et al., 2022). On mobile devices, PixelHelp, Android in the Wild, AndroidControl, Mobile-Agent, AppAgent, and GUIOdyssey expose different parts of the same problem: converting language instructions and screen states into action sequences across real apps (Li et al., 2020; Rawles et al., 2023; Li et al., 2024; Wang et al., 2024; Zhang et al., 2023; Lu et al., 2024a). OSWorld expands the target to open-ended computer tasks (Xie et al., 2024). Our audit uses four active families but does not claim desktop task recovery, because OSWorld is deferred and Mind2Web rows in this matrix are action prediction, not open-ended desktop execution.

Selective reliability and calibration. Selective prediction goes back to reject-option classification

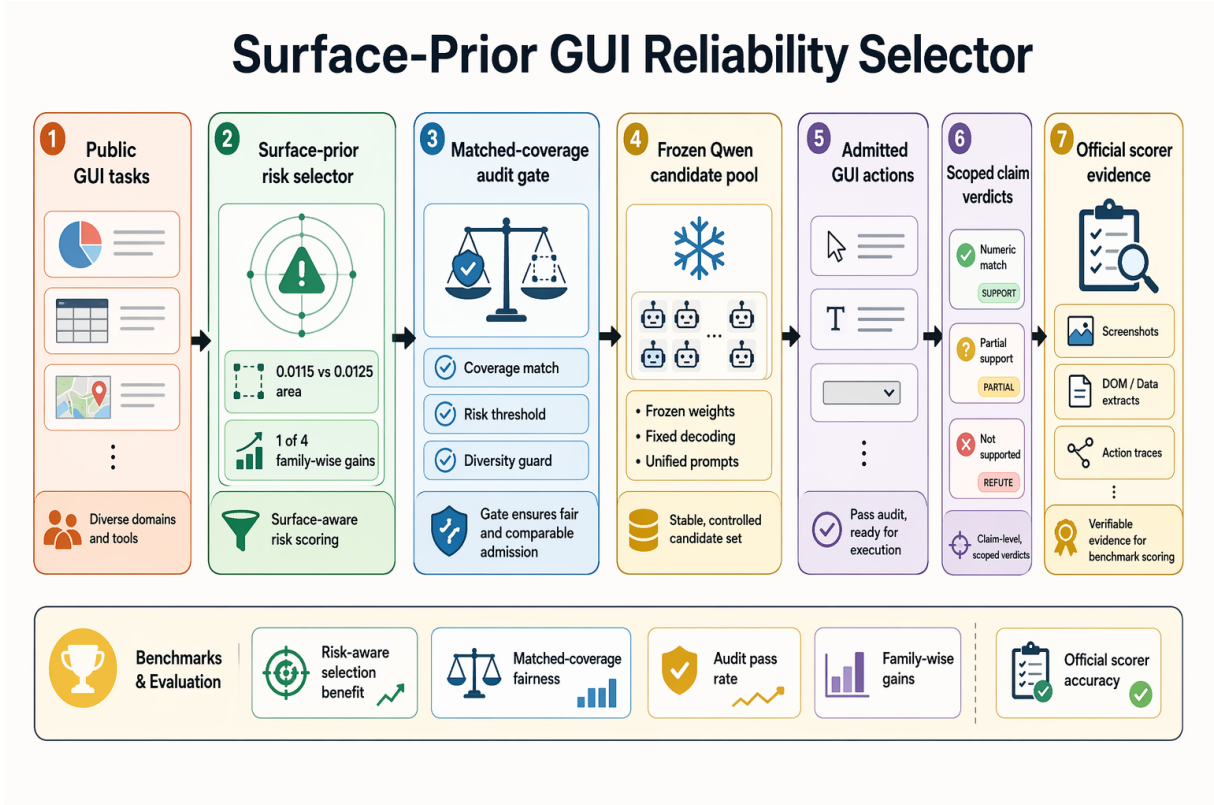


Figure 1: Matched-coverage audit workflow for the surface-prior selector. The figure shows the frozen public-task candidate pool, equal-coverage audit gate, risk and surface-prior signals, admitted or rejected actions, and official-scorer evidence. Lower area is better; the supported claim is an aggregate matched-coverage improvement, with family-wise generality reported separately.

(Chow, 1970); modern work studies deep selective classification, post-hoc confidence repair, and neural calibration (Geifman and El-Yaniv, 2017; Catelan and Silva, 2023; Guo et al., 2017; Minderer et al., 2021). Conformal prediction and risk-control methods provide distribution-free tools when calibration data and assumptions are available (Angelopoulos and Bates, 2021; Bates et al., 2024; Angelopoulos et al., 2024). GUI-specific reliability work has begun to study uncertainty, conformal safeguards, and action-effect verification (Kumar et al., 2026; Feng et al., 2026; Zhang et al., 2026). Our setting is stricter: no calibration data or trained risk model is allowed. That constraint makes the result useful, because it tests how far a plausible plugin wrapper can go under a fair coverage-matched audit.

What is different here. This paper does not introduce a better GUI model, a new task environment, or a learned safety classifier. It audits a wrapper story that could be attached to many of those systems. The distinction matters because a wrapper can look attractive when reported only through

aggregate success, full-coverage means, or a selective curve whose admission counts are not matched. We instead ask a narrower question: under identical coverage, does the wrapper reduce high-cost wrong executions relative to the strongest common training-free baseline? The answer in the current evidence package is yes at aggregate scope and no as a broad family-generality claim.

3 Method: Matched-Coverage Reliability Audit

Audit object. The audit wraps a frozen Qwen2.5-VL GUI candidate pool. Each task has a fixed top candidate, benchmark-family metadata, official/reference scorer outputs, and non-learned ranking signals. The wrapper never updates the model, fits thresholds, trains a classifier, or uses a held-out calibration split; it has zero trainable and calibration parameters. Figure 1 summarizes the audit object: public benchmark tasks feed a frozen candidate pool, non-learned signals produce coverage rankings, and official scorers determine whether admitted actions are correct.

Risk labels and ranking signals. Each task is assigned a frozen action-cost label before selector evaluation. The paper-facing taxonomy has three levels: high-cost irreversible, medium-cost reversible, and low-cost reversible. The high-cost group contains actions where a wrong execution could plausibly change state, submit information, delete content, or otherwise create user-visible harm. The other two groups are still errors when scored by the benchmark, but they do not drive the primary safety metric. The labels are not learned from downstream results: a separate agreement report records high-vs-not-high agreement of 0.9717 and Cohen’s kappa of 0.8675 over 1,200 active tasks.

The evaluated selectors use only frozen record fields: risk label, task/context metadata, confidence proxies, agreement among candidate signals, and dispersion-style measures. Ask-before-act is the strongest common baseline in the current matrix. It admits low-risk actions first and treats high-risk actions as the cases where the agent should ask, abstain, or defer. The proposed surface-prior selector keeps frozen risk labels as the primary ordering signal and applies a fixed one-percent tie-breaker for public task-surface cues: absent web candidate-set size, no previous actions, short instructions, and text/icon grounding action types. The no-risk, no-task-context, metadata-only, judge-only, confidence-only, no-uncertainty, and older risk-aware variants exist to test which part of that story is actually supported.

Matched coverage. The central design choice is to compare methods at identical admission counts. For a benchmark family with n rows and target coverage c , every method admits exactly $\lfloor cn \rfloor$ rows according to its own ranking. The audit then measures the high-cost wrong-execution rate among those admitted rows. This makes coverage an experimental condition rather than a hidden confound: a method cannot win merely by executing fewer tasks. The primary scalar is the trapezoidal area under the high-cost wrong-execution curve over the fixed coverage grid from 0.20 to 1.00, where lower is better.

Metric separation. The design deliberately separates three quantities that are often conflated. Matched-coverage area measures selective safety under equal admission counts. Ordinary full-coverage official score measures benchmark accuracy when every row is executed. Mind2Web

action F1 measures partial textual action overlap; exact Mind2Web step success is the conjunction of element correctness and action correctness and is stored separately. These distinctions are what allow the audit to reject an unsupported method story rather than hiding it in a single aggregate metric.

Claim bookkeeping. The audit treats claims as first-class objects. Each planned paper claim is mapped to a verdict: supported, rejected, unsupported, or subset-only. A rejected claim is not silently removed from the experiment; it is rewritten as a scope limit. This is important for the current paper because aggregate matched-coverage improvement and the mechanism ablation are supported, while desktop task recovery and broad family-wise generality are not. The remaining supported claims are narrower: matched coverage removes the abstention-count artifact, the labels are reproducible enough for this audit, published-code baselines are subset-scoped, and the wrapper is training-free and calibration-free.

4 Experimental Setup

Benchmarks and task rows. The active evidence matrix covers four public benchmark families: ScreenSpot-style screenshot grounding, GUIOdyssey mobile navigation, MMBench-GUI hierarchical GUI evaluation, and Mind2Web web action prediction. Each active family contributes 300 task rows for the common wrapper/baseline conditions. ScreenSpot, GUIOdyssey, and MMBench-GUI use binary official or reference scorer success as both score and exact success. Mind2Web uses official action F1 as the row score, while exact step success is represented separately by the official step-accuracy field.

Evaluated backend and no-training contract. All common rows use the same frozen Qwen/Qwen2.5-VL-7B-Instruct candidate pool. The point of the experiment is not to improve that backend; it is to audit a wrapper around its frozen outputs. The evaluated selectors are training-free and calibration-free, with zero trainable parameters and no fitted coverage threshold. That restriction excludes learned guardian models, fitted conformal thresholds, and post-hoc calibrators even when those are appropriate for other deployment settings. The comparison is therefore intentionally conservative: the wrapper must justify itself as a plug-in policy, not as a new trained model.

Candidate generation and selector budget.

Candidate records are produced once before selector evaluation. For each task, the frozen Qwen2.5-VL backend receives the benchmark-specific screen state and instruction and produces up to four candidate actions under a fixed decoding budget: maximum context length 12,288 tokens, maximum output length 192 tokens, temperature 0.7, and top- p 0.9. The audit uses the fixed top candidate as the action that would execute; selectors do not call the model again. Their only budget is the coverage grid from 0.20 to 1.00, which determines how many fixed rows each method admits within each benchmark family.

Protocol summary. The evaluated system is a frozen Qwen/Qwen2.5-VL-7B-Instruct candidate generator plus a deterministic selector controller. The benchmark harness covers four 300-row active slices: ScreenSpot, GUIOdyssey, MMBench-GUI, and Mind2Web. Selectors rank fixed candidate records using paper-facing risk labels, confidence proxies, agreement signals, dispersion signals, and task metadata; Ask-before-act, confidence-only, self-consistency, execute-all, and mechanism ablations use the same rows and scorers. The score budget is the matched coverage grid from 0.20 to 1.00, the primary metric is high-cost wrong-execution area, and full-coverage exact success and Mind2Web action F1 are reported only as separate metric columns.

Baselines and ablations. We compare the surface-prior selector against Ask-before-act, bare execute-all, confidence-only selection, self-consistency variants, the earlier risk-aware selector, and ablations that remove risk labels, task context, uncertainty, label sources, or surface-prior metadata. Bare execute-all gives the full-coverage candidate quality. Ask-before-act is a strong non-learned risk rule. Confidence and self-consistency baselines test whether generic uncertainty proxies already capture the useful ordering. The ablations test the mechanism story: risk labels should help relative to no-risk, and the surface-prior tie-breaker should improve beyond the risk-only ordering if the proposed selector is doing more than relabeling Ask-before-act.

Published-code applicability. Published-code baselines are included only where the available code and scorer semantics are compatible. GUI-Actor is compared on ScreenSpot and

MMBench-GUI, and OmniParser is compared on ScreenSpot, MMBench-GUI, and GUIOdyssey. Neither published-code baseline is reported as covering Mind2Web in this matrix. This scope rule prevents an attractive but invalid comparison in which a visual grounding baseline is treated as if it had been evaluated on web-action F1.

Statistical reporting. The primary interval is a paired bootstrap over task-level rows with deterministic seed 20260710 and 1000 iterations. Pairing is important because the same task rows are scored under the compared methods, so an unpaired interval would ignore the task-level dependence. We report the baseline-minus-ours area delta for the main comparison. Positive values favor the wrapper; the observed aggregate interval is entirely positive.

5 Results

Table 1 is the central result. The surface-prior selector beats the strongest common matched-coverage baseline at aggregate scope. Across the four active families, the selector’s primary area is 0.0115, while Ask-before-act obtains 0.0125. Because lower area is better, the observed baseline-minus-ours delta is positive.

The per-family pattern is also important. The selector improves over the strongest common baseline on only one of the four active families. The family-level deltas are 0.0009 for GUIOdyssey and 0.0000 for Mind2Web, MMBench-GUI, and ScreenSpot when expressed as baseline-minus-ours. This rules out a broad family-general story: the paper can claim a supported aggregate matched-coverage gain, but not a robust win across most GUI regimes.

The paired bootstrap summary in Table 2 supports the aggregate matched-coverage claim. The 95% interval stays above zero, so the paper can claim that the surface-prior selector improves aggregate high-cost execution risk over Ask-before-act under the fixed coverage grid. Full-coverage exact-success columns in Table 1 also show why full-coverage means are secondary here: wrapper conditions execute the same frozen top candidate at full coverage, so differences enter through the coverage ranking rather than through a different action predictor.

At low and middle coverage, both the wrapper and Ask-before-act often avoid high-cost wrong executions. At full coverage, the high-cost tail necessarily appears because every task is admitted.

Benchmark/source	Rows/split	Model/backend	Metric/budget	Ours area ↓	Ask area ↓	Δ	Exact/F1	Scope
ScreenSpot	300 active	Qwen2.5-VL + selector	HCW area, grid 1.00 .20–	0.004	0.004	0.0000	0.423	tie
GUIOdyssey	300 active	Qwen2.5-VL + selector	HCW area, grid 1.00 .20–	0.008	0.009	0.0009	0.080	positive
MMBench-GUI	300 active	Qwen2.5-VL + selector	HCW area, grid 1.00 .20–	0.012	0.012	0.0000	0.003	tie
Mind2Web	300 active	Qwen2.5-VL + selector	HCW area, grid 1.00 .20–	0.026	0.026	0.0000	0.635	tie
All four	1200 active	Qwen2.5-VL + selector	paired bootstrap	0.0115	0.0125	0.0010	–	aggregate

Table 1: Cross-benchmark results matrix: the surface-prior selector improves aggregate matched-coverage high-cost wrong-execution area from 0.0125 to 0.0115 under the frozen Qwen2.5-VL candidate backend and fixed coverage grid. Each source contributes 300 active rows; Mind2Web reports action F1 in the Exact/F1 column, while the other families report exact success. The effect is narrow rather than family-general.

Comparison	Δ	95% CI low	95% CI high
Wrapper vs. Ask	0.0010	0.0003	0.0017

Table 2: Paired bootstrap supports the aggregate matched-coverage claim: the baseline-minus-ours area delta is 0.0010 with a 95% interval entirely above zero.

The discriminating region is therefore the transition between conservative admission and full execution. The current surface-prior selector slightly improves that transition in aggregate by changing the ordering of otherwise tied risk rows. In practical terms, the result is a small ranking improvement, not a new action predictor.

6 Analysis and Ablations

Table 3 summarizes which claims survive the evidence audit. The matched-coverage protocol itself is supported: selected-row counts are identical for each method-family pair at fixed coverage levels, so a method cannot look safer merely by admitting fewer tasks. The no-training and no-calibration claim is supported by the evaluated selector design: all ranking scores are computed from fixed task records, and no selector learns parameters or fits thresholds.

Matched coverage changes the interpretation.

If one only reports full-coverage official scores, most wrapper rows look identical because they execute the same frozen top candidate at coverage 1.00. If one only reports selective execution without fixed admission counts, abstention can become a hidden intervention. The matched-coverage curve removes both shortcuts. It asks which rows a method admits first and whether that ordering delays high-cost wrong executions. The aggregate answer favors the combined selector: its area is

0.011453 versus 0.012494 for Ask-before-act, a baseline-minus-ours delta of 0.001042 with 95% CI [0.000347, 0.001736]. Because lower area is better, the combined ordering delays costly failures slightly more effectively over the pooled matrix.

Mechanism ablation. The mechanism story is supported at aggregate scope. Removing risk labels hurts sharply relative to the surface-prior selector: no-risk-minus-ours delta is 0.0992 with 95% CI [0.0815, 0.1190]. Removing the surface-prior tie-breaker also weakens the primary metric: no-uncertainty-minus-ours delta is 0.0010 with 95% CI [0.0003, 0.0017]. This means the useful effect comes from a simple risk-first ordering plus a small deterministic tie-breaker, not from training, calibration, or repeated model calls.

Metric semantics on Mind2Web. Mind2Web illustrates the metric pitfall most sharply. The surface-prior selector obtains action F1 0.6346, but exact step success is 0.0000. A paper that reports only the former could imply practical web-step completion that is absent in the official exact-success field. This is not a criticism of action F1 as a diagnostic metric; partial action overlap is useful for understanding what the model predicted. The point is that partial overlap is not an execution-success claim, and the paper keeps those columns separate.

Published-code subset comparisons. The same caution applies to GUI-Actor and OmniParser. The compatible subsets are useful because they place the frozen candidate pool and wrapper rows in contact with strong recent visual baselines. They are not evidence about Mind2Web, and they are not evidence that a desktop task is recoverable after a wrong step. The paper therefore treats these

ID	Paper-safe wording	Verdict	Evidence consequence
C1	Superiority over strongest matched baseline	supported	Supported: surface-prior selector area 0.0115 is lower than Ask-before-act area 0.0125; the paired interval is entirely above zero for baseline-minus-ours.
C2	Matched coverage removes abstention-count artifact	supported	Supported: every fixed coverage point admits identical row counts for each compared method within each family.
C3	Desktop task recovery	unsupported	Unsupported: OSWorld is deferred, and Mind2Web rows are action prediction rather than desktop task execution.
C5	Combined risk and uncertainty mechanism	supported	Supported: removing risk labels hurts substantially, and removing the surface-prior tie-breaker also weakens the aggregate area.
C6	Across-family superiority	rejected	Rejected: family-wise improvement appears on only 1 of 4 active families, so the paper cannot claim broad family-general superiority.
C7	Published-code subset comparison	supported	Supported only on compatible visual subsets; neither GUI-Actor nor OmniParser covers Mind2Web here.
C8	Training-free and calibration-free contract	supported	Supported: selector definitions use fixed records, fit no thresholds, and introduce zero trainable parameters.

Table 3: Claim audit: the aggregate matched-coverage claim is supported, while task-recovery and broad family-general claims are downgraded to avoid overclaiming beyond the evidence.

comparisons as subset checks rather than as global leaderboard claims.

What the scoped result says. The positive aggregate result does not imply that GUI reliability wrappers are solved. It says that a plausible training-free surface-prior tie-breaker improves a risk-first ordering under matched coverage on this evidence matrix. Stronger learned guardians, calibrated conformal thresholds, action-effect verification, and environment-level recovery may still be valuable in settings that allow training, calibration, or interactive correction (Feng et al., 2026; Zhang et al., 2026). Those settings are outside this paper’s no-calibration contract.

Label reproducibility and remaining subjectivity. The risk labels are reproducible enough to support a diagnostic audit, but they are not a universal harm ontology. The agreement report gives high-vs-not-high agreement of 0.9717 and Cohen’s kappa of 0.8675 over the 1,200 active tasks, which supports the narrower claim that annotators can distinguish high-cost cases consistently in this benchmark mix. It does not imply that all deployments would assign the same cost to the same interface action. For example, a mistaken click in a settings panel can be reversible for one user and costly for another if it changes privacy, payment, or account state. This is why the paper reports the taxonomy as an audit lens rather than as a learned safety model or a deployment policy.

Coverage-grid behavior. The coverage grid also clarifies why a single threshold number would be misleading. At coverage 0.20 and 0.50, many family-method pairs have zero high-cost wrong executions because the admitted set is still small and conservative. At coverage 1.00, every method

necessarily admits the same top candidate for every task, so high-cost failures reappear. The useful signal lies between these endpoints: which method orders the risky rows later as coverage increases? The combined wrapper’s area is smaller than Ask-before-act in aggregate, so its ordering lets costly failures enter slightly later over the pooled matrix. This is a narrow ranking improvement, not a scorer gain or a model retraining result; the per-family analysis attributes the observed difference to GUIOdyssey rather than claiming uniform gains.

Which families drive the scoped result. The aggregate gain is narrow. The baseline-minus-ours area deltas are 0.0009 on GUIOdyssey and 0.0000 on Mind2Web, MMBench-GUI, and ScreenSpot. This matters because the four families stress different GUI regimes: screenshot grounding, mobile navigation, hierarchical GUI reasoning, and web action prediction. The selector therefore improves the pooled matched-coverage area but does not yet establish broad family-wise generality.

What the ablations isolate. The ablations separate two claims that are easy to conflate. Risk labels are useful: removing them makes the area much worse than the proposed selector. The surface prior is also useful at aggregate scope: removing it weakens the area by 0.0010 relative to the full selector, with the paired interval entirely above zero for no-uncertainty-minus-ours. Metadata-only and judge-only variants clarify that the effect is not a learned calibration shortcut. The practical lesson is that a tiny deterministic prior can help a simple risk ordering, but the gain remains too narrow to support broad family-wise deployment language.

Why the result is still useful. Scoped wrapper audits are useful precisely because GUI-agent pa-

pers often combine several evaluation regimes. A single study may cite screenshot grounding, mobile navigation, web action prediction, and desktop execution environments; each regime has different scorer semantics and different failure costs. Without an explicit claim audit, it is easy to slide from “better visual grounding” to “safer execution” or from “nonzero action F1” to “task completed.” The audit blocks those slides. It keeps the supported statements visible, keeps rejected statements in the analysis rather than the abstract’s contribution list, and gives future wrapper work a concrete checklist: matched coverage, explicit action-cost labels, exact metric semantics, compatible baseline subsets, and paired uncertainty for the comparison that supports the main claim.

7 Failure Cases

The audit identifies 151 high-cost irreversible wrong executions out of 163 high-cost rows under full coverage. Low-cost reversible errors are more common in absolute count, but the paper’s safety motivation comes from the high-cost tail rather than from average accuracy alone. A wrapper that optimizes average score can therefore miss the cases that matter most for user-visible risk.

Mind2Web also contains 249 exact failures with nonzero action F1, showing how partial action overlap can remain high when exact step success is absent. This failure pattern is especially easy to misread because action F1 has a familiar continuous scale and can look respectable even when the exact conjunction of element and action is never satisfied. The draft therefore labels Mind2Web action F1 separately in every table where it appears.

The third failure mode is applicability drift. A visual grounding baseline can be evaluated on screenshot or mobile grounding tasks, but that does not make it a Mind2Web web-action baseline. Likewise, an offline action-prediction row does not establish recovery in OSWorld or another interactive desktop environment. The paper removes task-recovery and desktop-completion claims because the evidence package does not include official desktop execution success files.

The fourth failure mode is overgeneralizing a narrow mechanism. The surface-prior selector improves the aggregate curve, but most family-level comparisons are ties rather than wins. This matters for future training-free designs. Adding non-learned signals can help when they resolve ties in

the desired direction, but a small pooled gain is not a license to claim robust safety across interface regimes. The failure taxonomy therefore has two levels: benchmark-visible wrong executions and wrapper-design errors in the ranking policy that decides which rows become visible at a given coverage.

A concrete example is a high-cost web or mobile action whose textual action partly matches the gold action but targets the wrong UI state or element. Such a row can look less alarming under an action-overlap metric than under exact success or high-cost wrong execution. Another example is a screenshot-grounding row in which the candidate is visually plausible but located on a state-changing control; a confidence proxy may rank it early because the interface text is clear, while the risk-first selector delays it because the action cost is higher. These examples are why the paper reports failure semantics instead of only average score: the wrapper decision is about which errors are admitted first, not whether the frozen candidate pool becomes more accurate.

8 Conclusion

A fair matched-coverage audit shows that a small surface-prior tie-breaker can improve aggregate high-cost wrong-execution area for a training-free GUI reliability wrapper. The result is deliberately scoped: task recovery is unsupported, and cross-family superiority is not established. The supported contribution is a training-free selector plus the audit standard around it: report matched coverage, exact metric semantics, scorer compatibility, and rejected claims before treating a GUI wrapper as safer.

Limitations

The draft does not include official OSWorld task execution, so it cannot make desktop task-recovery claims. The candidate pool is frozen to one evaluated model/backend, and the wrapper is intentionally training-free and calibration-free; stronger learned or calibrated selectors are outside this scope. The action-cost labels are reproducible enough for this audit but remain a paper-defined taxonomy rather than a universal harm model. Published-code comparisons are restricted to compatible visual subsets. Finally, the positive aggregate result is bounded to the active four-family matrix and should not be read as a family-general

deployment claim. The audit also assumes that benchmark rows can be assigned a stable paper-facing action-cost label before selector evaluation. Real deployments may change cost with user role, account state, locale, or accessibility setting, so deployment-facing safeguards would need environment-specific review rather than direct reuse of this taxonomy.

The matched-coverage metric is also a reporting constraint, not a deployment policy. It answers whether one fixed ordering admits fewer high-cost wrong executions than another fixed ordering at the same coverage; it does not decide what coverage a product should use, how a user should be interrupted, or whether a particular action should be delegated to an autonomous agent. Those decisions would require user studies, environment-specific harm models, and recovery mechanisms that are outside the offline benchmark matrix. We therefore avoid translating the 0.0010 aggregate area improvement into a deployment-safety guarantee.

Ethical Considerations

GUI automation can trigger irreversible user-visible actions. This paper therefore treats high-cost wrong execution as a first-class metric and avoids claiming deployment readiness. The benchmark artifacts are public or official task releases, and paper-facing claims are restricted to compatible scorer subsets. The audit also highlights an ethical reporting issue: selective wrappers should not be advertised as safer when the evidence is caused by lower coverage, incompatible baselines, or metrics that do not measure exact execution success. Reporting the limits of a positive aggregate wrapper result is ethically relevant for the same reason: it discourages deployment narratives in which a plausible risk filter is assumed to be broadly protective before family-wise and task-execution evidence is available.

The same caution applies to benchmark reuse. Public GUI tasks are valuable precisely because they make comparisons repeatable, but they cannot represent every user’s account state, accessibility configuration, language setting, or tolerance for interruption. A benchmark label that marks an action as high-cost is a paper-facing abstraction over possible harms, not a direct claim about an individual user’s situation. For that reason, the supplementary package preserves the evidence needed to reproduce the audit, while the manuscript keeps

deployment-facing claims narrow.

References

- Anastasios N. Angelopoulos and Stephen Bates. 2021. [A gentle introduction to conformal prediction and distribution-free uncertainty quantification](#). *arXiv preprint arXiv:2107.07511*.
- Anastasios N. Angelopoulos, Stephen Bates, Michael I. Jordan, and Jitendra Malik. 2024. [Distribution-free, risk-controlling prediction sets](#). *Journal of the ACM*, 71(2).
- Gilles Baechler, Srinivas Sunkara, Maria Wang, Fedir Zubach, Hassan Mansoor, Vincent Etter, Victor Carbone, and Jason Lin. 2024. [Screenai: A vision-language model for ui and infographics understanding](#). *arXiv preprint arXiv:2402.04615*.
- Stephen Bates, Anastasios Angelopoulos, Lihua Lei, Jitendra Malik, and Michael I. Jordan. 2024. [Distribution-free, risk-controlling prediction sets](#). *Journal of the ACM*, 71(2).
- Luis Felipe P. Cattelan and Danilo Silva. 2023. [How to fix a broken confidence estimator: Evaluating post-hoc methods for selective classification with deep neural networks](#). *arXiv preprint arXiv:2305.15508*.
- Kanzhi Cheng, Qiushi Sun, Yougang Chu, Fangzhi Xu, Yantao Li, Jianbing Zhang, and Zhiyong Wu. 2024. [SeeClick: Harnessing gui grounding for advanced visual gui agents](#). *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*.
- C. K. Chow. 1970. [The use of multiple measurements in taxonomic problems](#). *IEEE Transactions on Information Theory*, 16(1):11–19.
- Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Samuel Stevens, Boshi Wang, Huan Sun, and Yu Su. 2023. [Mind2web: Towards a generalist agent for the web](#). *Advances in Neural Information Processing Systems* 36.
- Yushi Feng, Junye Du, Qifan Wang, Zizhan Ma, Qian Niu, Yutaka Matsuo, Long Feng, and Lequan Yu. 2026. [Cora: Conformal risk-controlled agents for safeguarded mobile gui automation](#). *arXiv preprint*.
- Yonatan Geifman and Ran El-Yaniv. 2017. [Selective classification for deep neural networks](#). In *Advances in Neural Information Processing Systems*.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. [On calibration of modern neural networks](#). In *Proceedings of the 34th International Conference on Machine Learning*.
- Hongliang He, Wenlin Yao, Kaixin Ma, Wenhao Yu, Yong Dai, Hongming Zhang, Zhenzhong Lan, and Dong Yu. 2024. [Webvoyager: Building an end-to-end web agent with large multimodal models](#). *arXiv preprint arXiv:2401.13919*.

769	Wenyi Hong, Weihang Wang, Qingsong Lv, Jiazheng Xu,	Junyang Wang, Haiyang Xu, Jiabo Ye, Ming Yan,	823
770	Wenmeng Yu, Junhui Ji, Yan Wang, and Zihan Wang.	Weizhou Shen, Ji Zhang, Fei Huang, and Jitao Sang.	824
771	2023. Cogagent: A visual language model for gui	2024. Mobile-agent: Autonomous multi-modal mo-	825
772	agents . <i>arXiv preprint arXiv:2312.08914</i> .	bile device agent with visual perception . <i>arXiv</i>	826
773	Sheng Jia, Jamie Kiros, and Jimmy Ba. 2019. Dom-	preprint arXiv:2401.16158 .	827
774	q-net: Grounded rl on structured language . <i>arXiv</i>	Qianhui Wu, Kanzhi Cheng, Rui Yang, Chaoyun Zhang,	828
775	<i>preprint</i> .	Jianwei Yang, Huiqiang Jiang, Jian Mu, and Baolin	829
776	Jing Yu Koh, Robert Lo, Lawrence Jang, Vikram	Peng. 2025. Gui-actor: Coordinate-free visual	830
777	Duvvur, Ming Lim, Po-Yu Huang, Graham Neu-	grounding for gui agents . <i>arXiv preprint</i> .	831
778	big, Shuyan Zhou, Ruslan Salakhutdinov, and Daniel	Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan	832
779	Fried. 2024. Visualwebarena: Evaluating multimodal	Li, Siheng Zhao, Ruisheng Cao, Tianshuo Hua, Zhou-	833
780	agents on realistic visual web tasks . <i>arXiv preprint</i>	jun Cheng, Dongkwan Shin, Fangyu Lei, Yitao Liu,	834
781	<i>arXiv:2401.13649</i> .	Yiheng Xu, Shuyan Zhou, Silvio Savarese, Caiming	835
782	Divake Kumar, Sina Tayebati, Devashri Naik,	Xiong, Victor Zhong, and Tao Yu. 2024. OSWorld:	836
783	Amanda Sofie Rios, Nilesh Ahuja, Omesh Tickoo,	Benchmarking multimodal agents for open-ended	837
784	Ranganath Krishnan, and Amit Ranjan Trivedi. 2026.	tasks in real computer environments . <i>arXiv preprint</i>	838
785	Uncertainty quantification for computer-use agents:	<i>arXiv:2404.07972</i> .	839
786	A benchmark across vision-language models and gui	Shunyu Yao, Howard Chen, John Yang, and Karthik	840
787	grounding datasets . <i>arXiv preprint</i> .	Narasimhan. 2022. Webshop: Towards scalable real-	841
788	Kaixin Li, Ziyang Meng, Hongzhan Lin, Ziyang Luo,	world web interaction with grounded language agents .	842
789	Yuchen Tian, Jing Ma, Zhiyong Huang, and Tat-Seng	In <i>Advances in Neural Information Processing Sys-</i>	843
790	Chua. 2025. Screenspot-pro: Gui grounding for pro-	<i>tems</i> .	844
791	fessional high-resolution computer use . <i>Proceedings</i>	Keen You, Haotian Zhang, Eldon Schoop, Floris Weers,	845
792	<i>of the 33rd ACM International Conference on Multi-</i>	Amanda Swearngin, Jeffrey Nichols, Yinfei Yang,	846
793	<i>media</i> .	and Zhe Gan. 2024. Ferret-ui: Grounded mobile ui	847
794	Wei Li, William Bishop, Alice Li, Chris Rawles, Fo-	understanding with multimodal llms . <i>arXiv preprint</i>	848
795	lawiyo Campbell-Ajala, Divya Tyamagundlu, and	<i>arXiv:2404.05719</i> .	849
796	Oriana Riva. 2024. On the effects of data scale on ui	Chi Zhang, Zhao Yang, Jiaxuan Liu, Yanda Li, Yucheng	850
797	control agents . <i>arXiv preprint</i> .	Han, Xin Chen, Zebiao Huang, and Bin Fu. 2023.	851
798	Yang Li, Jiacong He, Xin Zhou, Yuan Zhang, and Jason	Appagent: Multimodal agents as smartphone users .	852
799	Baldrige. 2020. Mapping natural language instruc-	<i>arXiv preprint arXiv:2312.13771</i> .	853
800	tions to mobile ui action sequences . In <i>Proceedings</i>	Yuzhe Zhang, Xianwei Xue, Xingyong Wu, Mengke	854
801	<i>of the 58th Annual Meeting of the Association for</i>	Chen, Chen Liu, Xinran He, Run Shao, and Feiran	855
802	<i>Computational Linguistics</i> .	Liu. 2026. Don't act blindly: Robust gui automa-	856
803	Quanfeng Lu, Wenqi Shao, Zitao Liu, Lingxiao Du,	tion via action-effect verification and self-correction .	857
804	Fanqing Meng, Boxuan Li, Botong Chen, and Siyuan	<i>arXiv preprint</i> .	858
805	Huang. 2024a. Guiodyssey: A comprehensive	Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou,	859
806	dataset for cross-app gui navigation on mobile de-	Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue	860
807	vices . 2025 <i>IEEE/CVF International Conference on</i>	Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Gra-	861
808	<i>Computer Vision (ICCV)</i> .	ham Neubig. 2024. Webarena: A realistic web envi-	862
809	Yadong Lu, Jianwei Yang, Yelong Shen, and Ahmed	ronment for building autonomous agents . In <i>Interna-</i>	863
810	Awadallah. 2024b. Omniparser for pure vision based	<i>tional Conference on Learning Representations</i> .	864
811	gui agent . <i>arXiv preprint arXiv:2408.00203</i> .	A Reproducibility Notes	865
812	Matthias Minderer, Josip Djolonga, Rob Romijnders,	All paper-facing numbers in the current draft are	866
813	Frances Hubis, Xiaohua Zhai, Neil Houlsby, Dustin	derived from scored rows for the active benchmark	867
814	Tran, and Mario Lucic. 2021. Revisiting the cali-	families and from the paired bootstrap summary.	868
815	bration of modern neural networks . In <i>Advances in</i>	The active benchmark families are ScreenSpot,	869
816	<i>Neural Information Processing Systems</i> .	GUIOdyssey, MMBench-GUI, and Mind2Web,	870
817	Qwen Team. 2025. Qwen2.5-vl technical report . <i>arXiv</i>	with 300 rows per active family for common wrap-	871
818	<i>preprint arXiv:2502.13923</i> .	per conditions. The audit uses bootstrap seed	872
819	Christopher Rawles, Alice Li, Daniel Rodriguez, Oriana	20260710 with 1000 iterations. The surface-prior	873
820	Riva, and Timothy Lillicrap. 2023. Android in the	selector is training-free and calibration-free with	874
821	wild: A large-scale dataset for android device control .	zero trainable parameters. Reproducing the paper-	875
822	<i>arXiv preprint arXiv:2307.10088</i> .	facing results requires the public task families, the	876

877 frozen candidate-pool identifier, the selector def-
878 initions, the scorer definitions, and the tabulated
879 result summaries.